

A Restricted Replication of Ottonello and Winberry (2020):

Financial Heterogeneity and the Investment Channel of Monetary Policy

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Abstract

This paper documents a replication of the main empirical results of Ottonello and Winberry (2020), who show that firms with lower default risk respond more strongly to monetary policy shocks. I translate the authors’ Stata replication code into R and reconstruct the quarterly Compustat firm panel, the high-frequency monetary shock series, and the baseline interaction regressions. The replication reproduces the paper’s core result: on the available post-1994 shock sample, the baseline coefficient on the leverage–shock interaction is -0.73 (s.e. 0.35), against the authors’ own post-1994 estimate of -0.71 (s.e. 0.35). The evidence suggests that the remaining discrepancies are driven mainly by differences in shock-series coverage. The replication excludes distance-to-default, Greenbook forecast, GSS shock-decomposition, and credit-rating specifications, and I document how every excluded column across the paper’s tables maps to one of these restrictions.

1 Introduction

Aggregate investment is among the most responsive components of GDP to monetary policy, yet firms differ enormously in their financial positions, and it is not obvious which firms drive this response. Ottonello and Winberry (2020, hence OW) address this question and reach a conclusion that runs against the conventional financial-accelerator intuition: firms with *low* default risk—low leverage and high distance to default—are the most responsive to monetary policy shocks. Their empirical result has become a benchmark in the literature on firm heterogeneity and monetary transmission, and their identification strategy—interacting

within-firm variation in financial position with high-frequency monetary shocks—has been widely adopted in subsequent work.

The influence of the paper makes it a natural candidate for replication. Replication serves two purposes here. First, it verifies that the published estimates can be recovered from the public replication package and standard data sources, and it makes explicit which results depend on data that are not freely available. Second, this replication is conducted in a different statistical environment from the original: the authors’ Stata pipeline is reconstructed in R. The cross-software reconstruction also provides a useful reproducibility check, because seemingly minute differences in software behavior can silently alter estimates. Documenting and resolving these differences is a central contribution of this report.

The replication is deliberately restricted in scope. In consultation with my supervisor, the exercise focuses on the leverage measure of financial position interacted with the baseline “wide-window” high-frequency monetary shock. Four components of the original analysis are excluded throughout: (i) specifications using distance to default, (ii) robustness checks using Greenbook forecast revisions, (iii) the decomposition of monetary shocks into target and path components following Gürkaynak, Sack, and Swanson (2005), and (iv) specifications using credit ratings. In addition, the extended 1972–1989 panel period is not used. The omitted results follow the documented data and scope restrictions.

The headline finding is that the replication succeeds. Because the monetary shock series available to this project begins in 1994 rather than 1990, the appropriate benchmark is the paper’s own post-1994 robustness estimates: against these, the replicated baseline coefficient is -0.73 (s.e. 0.35) versus a published -0.71 (s.e. 0.35), with the R^2 agreeing to within 0.001 and the sample size within 2% . Across the full set of in-scope tables, all replicated columns mostly match their Stata counterparts, with the differences attributable to the shorter shock sample rather than to the translated code.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the original paper’s research question, data, and empirical strategy. Section 4 documents the replication methodology, including the Stata-to-R translation and its limitations. Section 5 compares descriptive statistics, and Section 6 presents the table-by-table regression comparisons and the replication of Figure 1(a). Section 7 synthesizes the sources of remaining discrepancies, section 8 sets out six practical lessons from the translation that generalise to other cross-software replications, and Section 9 concludes.

2 Literature Review

Monetary policy affects firm investment by changing interest rates, financing costs, and the expected return to capital. However, these effects are not uniform across firms. Gertler and Gilchrist (1994) provide early evidence of firm heterogeneity, showing that smaller firms experience larger declines in sales and inventories following monetary tightening. Their findings helped motivate a broader literature on the role of financial frictions in monetary policy transmission. Subsequent studies have examined several different sources of firm-level heterogeneity. Durante, Ferrando, and Vermeulen (2022) find that younger and more highly leveraged firms respond more strongly to monetary policy, while firms producing durable goods display large and persistent investment responses. Cloyne et al. (2023) find that young firms that do not pay dividends adjust investment and borrowing more strongly following changes in interest rates. Jeenas (2026) instead emphasizes liquidity, finding that firms with lower cash balances reduce investment more strongly following monetary tightening. Albrizio, González, and Khametshin (2026) find that the marginal revenue product of capital (MRPK) predicts heterogeneous investment responses better than measures such as age, leverage, and liquidity. Drechsel et al. (2026) take a broader approach and estimate the distribution of firm investment responses without selecting a single firm characteristic in advance. They find that most firms respond only modestly, while a small share of firm-time observations respond strongly, and that standard firm characteristics explain only part of this heterogeneity.

Ottonello and Winberry (2020) differ from much of the literature on this topic. They find that firms with lower leverage and greater distance to default, indicating lower default risk, respond more strongly to monetary policy shocks. Their explanation is that severe financial frictions can restrict a firm’s ability to adjust investment, even when monetary conditions become more favorable. Their empirical specification also focuses on changes in financial position within firms, which distinguishes their result from studies that mainly compare different groups of firms. The present project provides a restricted replication of the empirical results in Ottonello and Winberry (2020). It compares the original descriptive statistics, regression tables, and Figure 1(a) with results produced from an R reconstruction of the authors’ Stata pipeline. The replication also documents differences caused by data availability and omitted variables, particularly distance to default, credit ratings, and Greenbook forecasts.

3 The Original Paper: Question, Data, and Empirical Strategy

3.1 Research Question

OW ask which firms are most responsive to monetary policy shocks. The answer is theoretically ambiguous. Financial frictions generate an upward-sloping marginal cost curve for financing investment, which dampens the response of constrained firms; but monetary policy may also flatten that curve—by raising cash flows or collateral values—which would amplify the response of constrained firms. The latter view, the financial-accelerator logic applied across firms, was the prevailing intuition. OW’s empirical contribution is to show that the data favor the former: firms with low default risk respond more strongly. This replication concerns only the empirical half of the paper.

3.2 Monetary Policy Shocks

OW measure monetary shocks with the high-frequency event-study approach. For an FOMC announcement at time t , the shock is

$$\varepsilon_t^m = \tau(t) \times (\text{ffr}_{t+\Delta_+} - \text{ffr}_{t-\Delta_-}), \quad (1)$$

where ffr is the implied Fed Funds rate from the current-month federal funds futures contract, the window is 15 minutes before to 45 minutes after the announcement, and $\tau(t)$ adjusts for the timing of the announcement within the month. The raw series runs from January 1990, when the federal funds futures market opened, to December 2007, restricting attention to conventional monetary policy; it contains 164 announcements. The high-frequency shocks are aggregated to quarterly frequency using a moving average weighted by the number of days remaining in the quarter after each announcement, so that shocks are weighted by the time firms have had to react to them.

3.3 Firm-Level Data

Firm variables come from quarterly Compustat, which OW describe as, to their knowledge, the only U.S. dataset that is simultaneously quarterly, a long panel, and rich in balance-sheet detail. The dependent variable is $\Delta \log k_{jt+1}$, the log change in the book value of firm j ’s tangible capital stock. The capital stock series is initialized from gross PPE and accumulated forward using net PPE changes. Financial position is measured as leverage ℓ_{jt} , the ratio of

total (short- plus long-term) debt to total assets. Investment, leverage, and distance to default are winsorized at the top and bottom 0.5%.

3.4 Baseline Specification

The baseline empirical specification is

$$\Delta \log k_{jt+1} = \alpha_j + \alpha_{st} + \beta (\ell_{jt-1} - \mathbb{E}_j[\ell_{jt}]) \varepsilon_t^m + \Gamma' Z_{jt-1} + e_{jt}, \quad (2)$$

where α_j is a firm fixed effect, α_{st} is a sector-by-quarter fixed effect, ε_t^m is the quarterly monetary shock (sign-normalized so that a positive shock is an interest-rate cut), and Z_{jt-1} is a vector of controls containing lagged leverage, total assets, sales growth, current assets as a share of total assets, a fiscal-quarter indicator, and the interaction of demeaned leverage with lagged GDP growth. Demeaned leverage is standardized over the sample, so β is interpreted as the differential investment semielasticity of a firm one standard deviation more leveraged than its own sample average. The within-firm demeaning $\ell_{jt-1} - \mathbb{E}_j[\ell_{jt}]$ is the paper’s key identification choice: it ensures the coefficient is driven by how a *given* firm responds when its financial position deviates from its own norm, rather than by permanent cross-firm heterogeneity in responsiveness. Standard errors are two-way clustered by firm and quarter.

The dynamics of the differential response are estimated with local projections,

$$\log k_{jt+h} - \log k_{jt} = \alpha_{jh} + \alpha_{sth} + \beta_h (\ell_{jt-1} - \mathbb{E}_j[\ell_{jt}]) \varepsilon_t^m + \Gamma_h' Z_{jt-1} + e_{jth}, \quad (3)$$

which underlie Figure 1(a) of the original paper.

3.5 Headline Empirical Findings

In OW’s Table III, the interaction coefficient is -0.69 (s.e. 0.29) without firm controls and -0.57 (s.e. 0.27) with firm controls: a firm one standard deviation more leveraged than usual has an investment semielasticity roughly 0.6 units lower. Against an average semielasticity of about 2 (their column (5)), this is an economically large—approximately one-quarter—difference in responsiveness. The local projections show the differential response by leverage peaking around four quarters after the shock and dissipating by twelve quarters.

4 Replication Methodology

4.1 Overview and Scope

The replication translates the authors’ publicly available Stata replication package into R. The pipeline is organized into two scripts mirroring the structure of the original code: a panel-construction script, which builds the Compustat firm panel, merges the monetary shock series and macroeconomic aggregates, and constructs the capital stock and control variables; and an estimation script, which produces the regression tables. The options were chosen to align the small-sample corrections and clustering conventions as closely as possible with `reghdfe`.

The replication is restricted in scope in six ways, agreed in advance with my supervisor:

1. **Distance to default.** All specifications using dd_{jt} are excluded. Constructing the measure follows the iterative Merton procedure described by Bharath and Shumway (2008), which solves for the market value and volatility of firm assets from observed equity values. It requires three inputs unavailable to this project: the CRSP daily stock file, the CRSP–Compustat Merge concordance linking Compustat firm identifiers to CRSP company identifiers, and a daily risk-free rate series. The authors additionally note that their implementation requires several days of computation.
2. **Greenbook forecasts.** Robustness checks controlling for the information channel of monetary policy using Greenbook forecast revisions are excluded.
3. **GSS shock decomposition.** The decomposition of announcements into target and path factors following Gürkaynak, Sack, and Swanson (2005) is excluded; only the baseline wide-window shock is used.
4. **Credit ratings.** Specifications and validation exercises using S&P long-term issue ratings are excluded.
5. **Sample period.** The extended 1972–1989 panel is not constructed; the replication uses the paper’s baseline modern sample.
6. **Structural model.** The paper’s heterogeneous-firm New Keynesian model, and all tables reporting calibrated parameters or model-generated moments—Tables IV, V, VI, VIII, XVI, and XVII, together with columns (2) and (4) of Table VII—are outside the scope of this project by agreement with my supervisor. These are excluded on different grounds from the five restrictions above: they could not be replicated without solving the model, rather than for want of an input series.

I document how every excluded column across the paper’s tables maps to one of these restrictions.

4.2 Data Limitations

The single most consequential data limitation is the coverage of the monetary shock series: the shock data available to this project begin in 1994, whereas OW’s series begins in January 1990 with the opening of the federal funds futures market. The missing 1990–1993 announcements include several large shocks from the early-1990s easing cycle. As documented in Section 7, this difference appears to account for nearly all of the gap in the directly comparable baseline specification..

4.3 Stata-to-R Translation Issues

Translating the pipeline surfaced several substantive differences in language semantics that, left unaddressed, produce silently divergent samples and estimates. I document the three most important; each was identified by tracing row-count discrepancies between intermediate Stata and R datasets.

Missing-value semantics in filters. Stata represents numeric missing values as larger than finite numbers, whereas R represents them as `NA`. As a result, logically similar-looking sample restrictions can behave differently depending on how they are written. I therefore handled missing values explicitly when translating the original filters and compared observation counts after the principal sample restrictions.

Uniqueness assertions. Stata’s construction code deduplicates multiple Compustat records per firm-quarter by retaining the final-update record, then asserts that the panel is unique by firm and quarter, halting if it is not. The authors note that this affected a single observation in their October 2017 data. The R translation applied the same retention rule but implemented the check as a printed diagnostic rather than an enforced condition; in the 2026 Compustat vintage used here, 32,564 firm-quarters remained duplicated, arising from firms whose fiscal-year changes place two report dates in the same calendar quarter. These were resolved by retaining the final-update record and, where that did not break the tie, the later report date, and by replacing the diagnostic with a hard stop. Because duplicated rows enter the winsorisation bounds, the within-firm means used for demeaning, and the standardisation, this affected estimates as well as sample counts.

Lag construction in unbalanced panels. Stata’s time-series operators (`L.`) respect declared panel gaps under `tsset`. A naive calendar-ordered lag in R assigns the previous *observed* row as the lag even across gaps in a firm’s quarterly spell. Lag construction for firm and macro variables was re-implemented to be gap-aware, returning `NA` where the preceding calendar quarter is not present in the panel.

Additional smaller correspondences were required: converting `Inf` values arising from log transformations to `NA` prior to winsorization (Stata’s `winsor` never encounters `Inf`); matching Stata’s percentile definition by using `quantile(type = 2)` in R for descriptive statistics; and treating placeholder zeros in the ratings data as missing.

5 Descriptive Statistics

5.1 Monetary Shock Summary Statistics (Table 1)

Table 1 compares the summary statistics of the monetary policy shocks in the original paper (OW, Table I) with those produced by the replication pipeline. The original series covers announcements from January 1990 through December 2007; the shock data available to this project begin in 1994, so the comparison also serves as the first quantitative characterization of the data-coverage restriction discussed in Section 4.

Table 1: Summary statistics of monetary policy shocks: original vs. replication

	High frequency		Smoothed		Sum	
	OW	Repl.	OW	Repl.	OW	Repl.
Mean	−0.0185	−0.010	−0.0429	−0.024	−0.0421	−0.022
Median	0	0.000	−0.0127	−0.011	−0.00509	−0.005
S.D.	0.0855	0.084	0.108	0.086	0.124	0.108
Min	−0.463	−0.463	−0.480	−0.356	−0.479	−0.431
Max	0.152	0.152	0.233	0.233	0.261	0.261
Observations	164	118	71	55	72	56

Notes: “OW” columns reproduce Table I of Ottonello and Winberry (2020), covering 1/1/1990–12/31/2007. “Repl.” columns are computed from the shock series available to this project, which begins in 1994 and ends in December 2007. “High frequency” shocks are the announcement-level surprises from equation (1); “Smoothed” shocks are aggregated to quarterly frequency using the day-weighted moving average; “Sum” aggregates by summing all shocks within a quarter.

The main difference is the observation count: the shock series used here begins in 1994 rather than 1990, so the replication has 46 fewer announcements and 16 fewer quarterly observations in both the smoothed and summed series. Where the underlying shocks are

observed, the pipeline matches the original: the maxima match in all three columns and the high-frequency minimum matches at -0.463 . The replicated means are closer to zero because the missing early-1990s announcements were mostly expansionary surprises, a difference that carries through to the regression results in Section 7.

5.2 Firm-Level Summary Statistics (Table 2)

OW’s Table II has three panels: (a) marginal distributions of investment, leverage, distance to default, and credit rating; and (b)–(c) correlation matrices, raw and residualized, among leverage, distance to default, and credit rating. Only panel (a) is replicated, and only its investment and leverage columns: distance to default and credit ratings are excluded under restrictions (1) and (4), which rules out the remaining columns of panel (a) and panels (b)–(c) entirely, since those panels exist to validate the excluded proxies. Table 2 reports the comparison.

Table 2: Summary statistics of firm-level variables (OW Table II, panel (a)): original vs. replication

	$\Delta \log k_{jt+1}$		ℓ_{jt}	
	OW	Repl.	OW	Repl.
Mean	0.004	0.004	0.263	0.263
Median	−0.004	−0.004	0.201	0.202
S.D.	0.091	0.091	0.348	0.346
95th Percentile	0.128	0.127	0.719	0.717
Observations	343,276	342,061	343,276	342,061

Notes: “OW” columns reproduce panel (a) of Table II in Ottonello and Winberry (2020) for the period 1983q3–2014q4. Statistics are computed before winsorization. The distance-to-default and credit-rating columns are excluded from the replication (restrictions (1) and (4)).

The two distributions are close to identical: almost every statistic matches to the third decimal, and the replication contains 1,215 fewer firm-quarter observations, a difference of 0.35%. Since this table is constructed before the monetary shock merge, the 1994 coverage restriction plays no role here; the residual gap reflects the more recent Compustat vintage used in this project, which both adds and removes historical firm-quarters relative to the authors’ original extraction through restatements, delistings, and reclassifications.

6 Main Regression Results

Each subsection below presents the original table alongside its replication. Columns are matched one-to-one where in scope; excluded columns are annotated with the scope restriction from Section 4 that accounts for them.

6.1 Baseline and Post-1994 Benchmark (Tables III and XIV)

Table III of the main paper reports the baseline specification (2); Table XIV of the supplementary material re-estimates it using only post-1994 data. In the replication these two exercises coincide. Because the available monetary shock series begins in 1994, all firm-quarter observations before that date have a missing shock and are dropped from every regression automatically; imposing the post-1994 restriction explicitly therefore removes no additional observations, and the replicated Table XIV column (1) is numerically the same estimate as the replicated Table III column (2). The two tables are presented together for that reason. In the original the two differ substantially in sample size (219,402 versus 174,274 observations), so the collapse is itself a direct consequence of the coverage restriction.

Of the original columns, (1) and (2) of Table III and column (1) of Table XIV are in scope. Column (5) of Table III is re-estimated in modified form without the distance-to-default interaction and is therefore reported only for completeness. Columns (3) and (4) of Table III and columns (2)-(6) of Table XIV involve distance to default (restriction R1) or Greenbook forecast controls (restriction R2).

Against the like-for-like benchmark the replication is close to exact. The specification with firm controls yields -0.73 with a standard error of 0.35 , against OW’s post-1994 estimate of -0.71 with a standard error of 0.35 ; the R^2 agrees to three decimal places (0.139 versus 0.138). The difference in point estimates, 0.02 , is a small fraction of a standard error. The sample is $3,024$ observations smaller than the original post-1994 sample, a difference of 1.7% corresponding to incomplete shock coverage in the opening quarter of 1994.

The comparison against the full-sample estimate of -0.57 reflects the same restriction rather than a divergence in construction. OW’s own results establish the direction and size of the effect: moving from the 1990–2007 sample to the post-1994 sample changes the coefficient from -0.57 to -0.71 , an increase in magnitude of roughly one quarter that is nonetheless well within one standard error. The replication reproduces the post-1994 value. Because the specification without firm controls has no published post-1994 counterpart, its replicated value of -0.73 can only be compared to the full-sample -0.69 ; the close agreement between the replicated columns (1) and (2) does, however, reproduce OW’s observation that adding firm-level controls leaves the point estimate essentially unchanged.

Table 3: Baseline heterogeneous responses of investment to monetary policy

	Original			Replication	
	III (1) 1990–2007	III (2) 1990–2007	XIV (1) post-1994	(1) post-1994	(2) post-1994
leverage $\times$ ffr shock	−0.69 (0.29)	−0.57 (0.27)	−0.71 (0.35)	−0.73 (0.36)	−0.73 (0.35)
Observations	219,402	219,402	174,274	171,250	171,250
R^2	0.113	0.124	0.138	0.127	0.139
Firm controls	no	yes	yes	no	yes
Time sector FE	yes	yes	yes	yes	yes
Time clustering	yes	yes	yes	yes	yes

Notes: Dependent variable is $\Delta \log k_{jt+1}$. “Original” columns reproduce columns (1) and (2) of Table III in the main paper and column (1) of Table XIV in the supplementary material to Ottonello and Winberry (2020). Replication columns (1) and (2) correspond to Table III columns (1) and (2); replication column (2) is also the replicated Table XIV column (1), since the post-1994 restriction is not binding on a sample whose shock coverage already begins in 1994. Leverage is winsorized, demeaned within firm, and standardized over the sample; the shock is sign-normalized so that a positive value is an interest rate cut. All specifications include firm and sector-by-quarter fixed effects and a fiscal-quarter dummy, and control for the interaction of demeaned leverage with lagged GDP growth. Firm controls are lagged sales growth, size, and current assets as a share of total assets. Standard errors two-way clustered by firm and quarter.

Column (5) of Table III removes the sector-by-quarter fixed effects in order to estimate the average response to a monetary shock alongside the differential response:

$$\Delta \log k_{jt+1} = \alpha_j + \alpha_{sq} + \gamma \varepsilon_t^m + \beta (\ell_{jt-1} - \mathbb{E}_j[\ell_{jt}]) \varepsilon_t^m + \Gamma'_1 Z_{jt-1} + \Gamma'_2 Y_{t-1} + e_{jt}, \quad (4)$$

where α_{sq} is a sector-by-fiscal-quarter seasonal fixed effect and Y_t contains four lags each of GDP growth, the inflation rate, and the unemployment rate. The coefficient γ measures the average investment semielasticity, which OW estimate at roughly 2; it provides the scale against which the interaction coefficient β is economically interpreted. The original column includes both the leverage and distance-to-default interactions and is estimated on the smaller distance-to-default sample ($N = 119,750$); the replicated version includes only the leverage interaction (restriction R1) and is estimated on the leverage sample ($N = 130,745$). The estimated average semielasticity is 1.41 (0.94) against the original 2.14 (0.61), and the leverage interaction is -1.01 (0.37) against the original -0.14 (0.58). Because the specification and the estimation sample both differ, this column is reported for completeness rather than as a direct comparison; the original interaction coefficient is conditional on distance to default, which the replicated version omits.

6.2 Identification and Control-Set Robustness (Tables IX, X, XV)

Tables IX, X, and XV, reported in the supplementary material to Ottonello and Winberry (2020), test the robustness of the baseline result to changes in the empirical specification rather than in the sample or the shock series. None of the three has a published post-1994 counterpart, so each is benchmarked against OW’s full-sample estimate. As established in Section 6, restricting to post-1994 data moves the baseline coefficient from -0.57 to -0.71 , roughly a quarter larger in magnitude and well within one standard error; differences of comparable size below should be read as a consequence of that same restriction rather than as evidence against the specification.

6.2.1 Not Demeaning Financial Position (Table IX)

Table IX replaces the within-firm demeaned interaction $(\ell_{jt-1} - \mathbb{E}_j[\ell_{jt}])\varepsilon_t^m$ with the level interaction $\ell_{jt-1}\varepsilon_t^m$, standardized over the full sample. The exercise is the counterfactual behind OW’s identification argument in their Online Appendix A.1: without demeaning, the estimate absorbs permanent cross-firm differences in responsiveness as well as the within-firm variation the model speaks to. Columns (1) and (2) are in scope; columns (3) and (5) use credit ratings (restriction R4) and columns (4), (6), and (7) use distance to default (restriction R1). Column (7) is additionally replicated in modified form, without the distance-to-default interaction.

Table 4: Heterogeneous responses, not demeaning financial position (Table IX)

	Original		Replication	
	(1)	(2)	(1)	(2)
leverage $\times$ ffr shock	−1.00 (0.32)	−0.80 (0.29)	−0.94 (0.43)	−0.90 (0.37)
Observations	219,402	219,402	171,250	171,250
R^2	0.113	0.124	0.127	0.139
Firm controls	no	yes	no	yes
Time sector FE	yes	yes	yes	yes
Time clustering	yes	yes	yes	yes

Notes: Leverage is winsorized and standardized over the entire sample but *not* demeaned within firm. All other elements of the specification follow Table 3. Original columns (3)–(7) are excluded under restrictions R4 (credit ratings) and R1 (distance to default). Standard errors two-way clustered by firm and quarter.

Both in-scope columns replicate closely. Without firm controls the coefficient is -0.94 against a published -1.00 ; with firm controls it is -0.90 against -0.80 . Both differences

are well inside one standard error, and the second is in the direction and of the magnitude implied by the post-1994 restriction. The replication also reproduces the qualitative point of the exercise: the undemeaned coefficients are substantially larger in magnitude than their demeaned counterparts in Table 3 (-0.94 and -0.90 against -0.73 and -0.73), consistent with OW’s argument that the level specification partly captures permanent heterogeneity in responsiveness across firms rather than the within-firm variation of interest. The R^2 is essentially unchanged between the two specifications (0.139 in both), indicating that the demeaning choice affects the interpretation of the coefficient rather than the fit of the regression.

Column (7) of the original removes the sector-by-quarter fixed effects and includes both the leverage and distance-to-default interactions on the distance-to-default sample. The replicated version includes only the leverage interaction and is estimated on the leverage sample, giving -0.75 (0.42) on the interaction and 1.38 (0.94) on the shock level, against -0.71 (0.45) and 2.08 (0.59) in the original. As with column (5) of Table III, the specification and sample both differ and the column is reported for completeness rather than as a direct comparison.

6.2.2 Not Controlling for Differences in Cyclical Sensitivities (Table X)

Table X drops the interaction of demeaned financial position with lagged GDP growth from the control vector Z_{jt-1} . Only column (1) is in scope; columns (2) and (3) use distance to default (restriction R1), and column (4), which includes both interactions, is replicated in modified form without the distance-to-default term.

Table 5: Main results, not controlling for differences in cyclical sensitivities (Table X)

	Original (1)	Replication (1)
leverage $\times$ ffr shock	-0.30 (0.24)	-0.47 (0.35)
Observations	219,402	171,250
R^2	0.124	0.139
Firm controls	yes	yes
Time sector FE	yes	yes
Time clustering	yes	yes

Notes: As Table 3, except that Z_{jt-1} excludes the interaction of demeaned leverage with lagged GDP growth. Original columns (2)–(4) excluded under restriction R1. Standard errors two-way clustered by firm and quarter.

The replicated coefficient is -0.47 (0.35) against a published -0.30 (0.24), a difference of about half a standard error. More importantly, the replication reproduces the substantive

result: relative to its own baseline the coefficient falls in both cases. (from -0.57 to -0.30 in the original, from -0.73 to -0.47 here) and becomes statistically insignificant in both. This supports OW’s argument that controlling for differences in cyclical sensitivity across firms strengthens rather than manufactures the estimated heterogeneity, since the largest shocks in the sample coincide with the onset of recessions.

Column (4) of the original removes the sector-by-quarter fixed effects and includes both interactions on the distance-to-default sample; the modified replication gives -0.88 (0.37) on the leverage interaction and 1.42 (0.94) on the shock level, against -0.07 (0.51) and 2.14 (0.61). As elsewhere, the specification and sample differ and the column is not a direct comparison.

6.2.3 Controlling for Lagged Investment (Table XV)

Table XV adds the lagged change in the capital stock, $\Delta \log k_{jt}$, to the baseline specification. Only column (1) is in scope; columns (2) and (3) use distance to default (restriction R1), and column (4), which includes both interactions and removes the sector-by-quarter fixed effects, is replicated in modified form.

Table 6: Controlling for lagged investment (Table XV)

	Original (1)	Replication (1)
leverage $\times$ ffr shock	-0.39 (0.27)	-0.53 (0.36)
$\Delta \log k_{jt}$	0.20 (0.01)	0.203 (0.008)
Observations	219,402	171,250
R^2	0.159	0.174
Firm controls	yes	yes
Time sector FE	yes	yes
Time clustering	yes	yes

Notes: As Table 3, with the lagged change in log capital added to the right-hand side. Original columns (2)–(4) excluded under restriction R1. Standard errors two-way clustered by firm and quarter.

The coefficient on lagged investment replicates closely, at 0.203 (0.008) against a published 0.20 (0.01). Because this coefficient does not directly interact with the monetary shock, the close result provides a useful additional consistency check on the construction of capital and investment. It is nevertheless estimated on the same post-1994 regression sample and is therefore not completely independent of the sample restriction.

The interaction coefficient is -0.53 (0.36) against a published -0.39 (0.27), a difference of roughly one third of a standard error. The attenuation relative to the respective baselines is also comparable: adding lagged investment moves the coefficient from -0.57 to -0.39 in the original (68% of the baseline) and from -0.73 to -0.53 in the replication (73%). The replication therefore reproduces both the level and the direction of the adjustment.

Column (4) of the original removes the sector-by-quarter fixed effects and includes the distance-to-default interaction; the modified replication gives -0.77 (0.39) on the leverage interaction and 0.189 (0.009) on lagged investment, against -0.07 (0.59) and 0.15 (0.01). The original does not report the coefficient on the shock level for this column.

The three tables in this group replicate the original consistently. In each case the in-scope coefficient lies within one standard error of its published counterpart, and in each case the change relative to the baseline — larger without demeaning, roughly halved without cyclical controls, moderately attenuated with lagged investment — reproduces in both direction and approximate magnitude.

6.3 Expansionary vs. Contractionary Shocks (Table XI)

Table XI of the supplementary material splits the monetary shock into positive (expansionary) and negative (contractionary) components, each interacted with demeaned leverage. Columns (1) and (2) are in scope; columns (3) and (4) use distance to default (restriction R1). Column (1) is the same regression as the baseline, so the new content is the split in column (2).

The expansionary-shock coefficient replicates closely: -0.56 (0.40) against a published -0.61 (0.28). The contractionary-shock coefficient does not: -1.18 (0.83) against -0.44 (0.93), and the two point estimates now differ in relative size — the replication finds a larger contractionary response, the original finds a larger expansionary one.

This difference should be weighed against the significance of the underlying estimates. Neither contractionary coefficient is significant at the 10% level in either the original or the replication ($t = -0.47$ and $t = -1.42$, respectively), and the gap between the two point estimates (0.74) is smaller than either standard error. This is the only specification in which the relative ordering of the two coefficients reverses. However, neither contractionary coefficient is statistically significant, and the estimates are too imprecise individually to establish a meaningful difference. OW note the same imprecision in their own contractionary estimate and do not emphasize it for that reason; the replication is consistent with treating this split as inconclusive rather than as evidence against expansionary shocks driving the result.

Table 7: Expansionary vs. contractionary shocks (Table XI)

	Original		Replication	
	(1)	(2)	(1)	(2)
leverage $\times$ ffr shock	−0.57 (0.27)		−0.73 (0.35)	
leverage $\times$ pos. ffr shock		−0.61 (0.28)		−0.56 (0.40)
leverage $\times$ neg. ffr shock		−0.44 (0.93)		−1.18 (0.83)
Observations	219,402	219,402	171,250	171,250
R^2	0.124	0.124	0.139	0.139
Firm controls	yes	yes	yes	yes
Time sector FE	yes	yes	yes	yes
Time clustering	yes	yes	yes	yes

Notes: As Table 3. A positive shock corresponds to an interest rate cut, so “pos. ffr shock” is an expansionary surprise. Original columns (3)–(4) excluded under restriction R1. Standard errors two-way clustered by firm and quarter.

6.4 Tables Not Replicated (Tables XII and XIII)

Tables XII and XIII control for the information channel of monetary policy by including forecast revisions of output growth, inflation, and unemployment from FOMC Greenbooks. Every column of both tables conditions on these forecast revisions, so neither table has an in-scope subset: the exclusion is total rather than partial, and follows directly from restriction R2 (Section 4). Greenbook data are released with a five-year lag and were not obtained for this project.

6.5 Comparison with the Model: Data Columns (Table VII)

Table VII of the main paper reports the baseline estimate alongside the same regression on simulated model data. Columns (2) and (4), the model estimates, are outside this project’s scope. Column (1) reports the same core standardized-leverage specification as Table III column (2); column (3) reports the same coefficient without standardizing leverage.

The non-standardized coefficient is -3.70 (1.79) against a published -3.12 , about 19% larger, consistent with the gap in the standardized columns. The identical R^2 across both replication columns confirms the two specifications differ only in scaling.

As a check, the ratio of the two coefficients recovers the standard deviation of demeaned leverage: 0.196 here versus 0.183 in the original, close to the directly measured 0.193. This confirms the leverage variable itself is constructed correctly.

Table 8: Empirical results, data columns (Table VII)

	Original		Replication	
	(1) Std.	(3) Not std.	(1) Std.	(3) Not std.
demeaned leverage $\times$ ffr shock	−0.57 (0.29)	−3.12 (1.47)	−0.73 (0.35)	−3.70 (1.79)
Observations	219,402	219,402	171,250	171,250
R^2	0.12	0.12	0.139	0.139
Firm controls	yes	yes	yes	yes
Time FE	yes	yes	yes	yes

Notes: Columns labelled “Std.” standardize demeaned leverage over the sample; columns labelled “Not std.” do not. Both replication columns retain the standardized interaction of demeaned leverage with lagged GDP growth as a control, following the original code. Columns (2) and (4) of the original report estimates on model-simulated data and are outside the scope of this replication. Standard errors two-way clustered by firm and quarter.

6.6 Dynamics of the Differential Response (Figure 1(a))

Figure 1 of the main paper reports the local projection (3) at horizons $h = 0, \dots, 12$. Panel (a), which uses leverage, is replicated; panel (b) uses distance to default and is excluded under restriction R1. Figure 1 overlays the replicated path against the authors’ own estimates, obtained directly from the coefficients and standard errors in their replication output rather than read from the published figure.

The impact estimate has the same sign and is reasonably close relative to its standard error: $\beta_0 = -0.73$ in the replication against -0.57 published. Both paths deepen over the first several quarters and recover toward zero from around $h = 9$ onward, converging almost exactly by $h = 10$ – 11 ; the two paths cross near zero at $h = 12$. This reproduces OW’s stated findings that the peak difference occurs around four quarters after the shock and that the differences disappear roughly twelve quarters out.

The main difference is amplitude at the trough: the replicated path bottoms out near $h = 3$ – 4 at roughly twice as deep as the published path, before converging to it from $h = 5$ onward. The published path lies inside the replication’s 90% band at every horizon, and for much of the range the replication’s band is the wider of the two, reflecting the smaller and more concentrated set of monetary shocks available after 1994. The larger trough may partly reflect sensitivity to the shorter post-1994 shock sample. However, because the original paper does not report a comparable post-1994 local projection, the source of the remaining difference in magnitude cannot be determined conclusively.

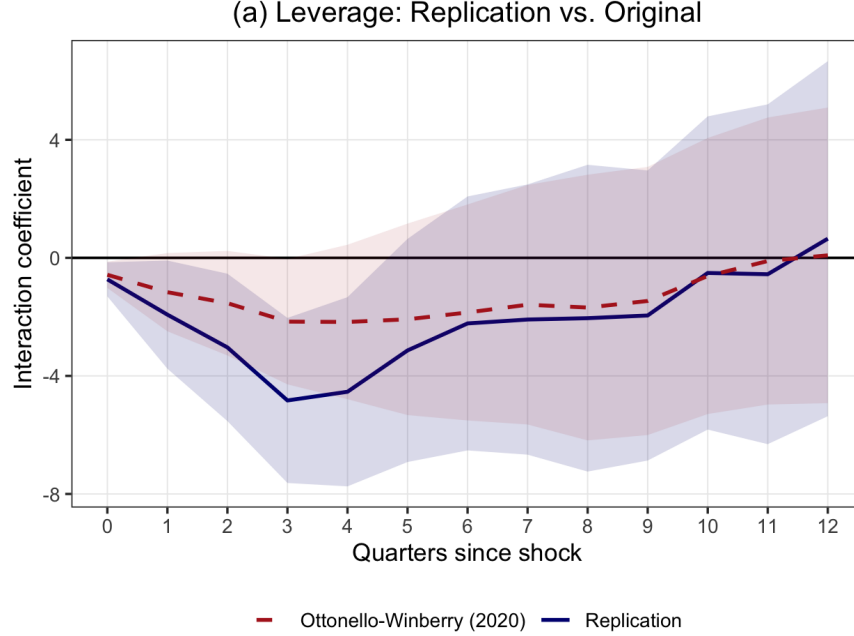


Figure 1: Dynamics of the differential response to monetary shocks, leverage: replication vs. original. Reports β_h from the local projection (3) over quarters h since the shock. Shaded regions are 90% error bands, two-way clustered by firm and quarter.

7 Discussion of Discrepancies

The replication’s central claim is that the remaining numerical differences between the R output and the published tables are attributable to the data available to this project rather than to the translated code. This section consolidates the evidence for that claim and identifies the two places where it requires qualification.

Shock coverage is the dominant source of difference. The monetary shock series available here begins in 1994Q2 rather than January 1990, removing roughly a fifth of the quarters that identify the interaction coefficient. The consequences are visible throughout: the replicated baseline is -0.73 against a published full-sample -0.57 , the replicated shock-dependent estimates are often larger in magnitude than their full-sample counterparts. That this reflects sample coverage rather than construction is established directly by the authors’ own post-1994 estimates, which are the correct like-for-like benchmark. Against those, the replicated baseline of -0.73 (s.e. 0.35) compares with -0.71 (s.e. 0.35), the R^2 agrees to within 0.001, and the sample is 1.7% smaller, corresponding to the absence of 1994Q1 from the shock series.

Quantities identified from the cross-section replicate more closely than quantities identified from the shock series. This distinction organizes the results. The coefficient on lagged

investment in Table XV — identified from the full panel of firm-quarters rather than from time-series variation in shocks — replicates at 0.203 against 0.20. The implied standard deviation of demeaned leverage recovered from Table VII is 0.196 against 0.183. The descriptive moments of investment and leverage in Table 2 match to within 0.002. By contrast, every coefficient that depends on the shock series is correctly signed. The pattern is consistent with a close reconstruction of the firm panel and estimates on a shorter, more concentrated set of monetary surprises.

Two results are only partly replicated. First, Table XI does not reproduce the original ordering of the expansionary and contractionary point estimates. Both interaction coefficients remain negative, consistent with low-leverage firms responding more strongly to monetary policy. However, whereas the original reports a larger coefficient in absolute value for expansionary shocks, the replication reports a larger coefficient for contractionary shocks. Neither coefficient in the replicated split specification is statistically significant at the 10% level, and no formal test of equality between the two coefficients is conducted. Table XI should therefore be interpreted as inconclusive about asymmetry, rather than as contradicting the paper’s main result.

Second, the local projection in Figure 1(a) reproduces the shape and timing of the published dynamics but not their amplitude, with a trough roughly twice as deep. Re-estimating the same projection on subsets of the post-1994 window moves the trough from -4.5 to -8.5 depending on which years are included—a range considerably wider than the gap between the replicated and published paths. The dynamics at longer horizons are therefore not precisely identified using 55 quarterly shocks, half of whose squared variation is concentrated in a single year. Unlike the baseline coefficient, these horizons have no published post-1994 benchmark against which the replication can be compared.

8 Lessons for Cross-Software Replication

Section 7 reports which differences remain and why. This section is about the process that got there: how the translation was checked, what went wrong along the way, and which of those experiences would transfer to another replication. Little of it is specific to this paper. The exercise was undertaken in part as a training exercise, and the six lessons below are the ones I would want to know at the start of a comparable project—particularly one conducted in a different language from the original, or without access to all of the original inputs.

1. Check the panel before checking the coefficient. The instinct when a replicated coefficient differs from a published one is to re-run the regression with a control added or

removed and see which change closes the gap. This is the wrong place to start. Leverage is winsorized at sample-wide cutoffs, demeaned within firm and standardized by a sample-wide standard deviation, so anything that changes which rows are in the panel changes the interaction regressor for every firm—including firms whose own records are perfectly correct. Comparing coefficients on a panel of unverified integrity therefore cannot distinguish a specification error from a sample error. I worked instead in order of severity: first, is the panel unique by firm and quarter; then, do the intended rows survive the trimming sequence; then, do the constructed variables have the right distributions; only then, do the estimates agree. The first of those tests found 32,564 duplicated firm-quarters, and every comparison made before it would have been meaningless.

Where the test is run matters as much as when. The obvious check on the merge that caused those duplicates is whether the join key is unique in the source file, and it was—that test alone would have closed the investigation. The defect lay in an intermediate bridge table, in which 1,295 firms mapped to more than one security identifier because firms change identifiers over time. The decisive test was on the object I was about to estimate on, not on the input I suspected.

2. Translate what a command does instead of what it looks like. The commands that caused trouble were the ones with an apparent one-line equivalent in the other language. Stata’s `merge m:1` is not a description of a merge but a condition imposed on it, halting if the condition fails, whereas R’s default join executes a fan-out silently; supplying the cardinality constraint explicitly restores the assertion. Because comparisons involving NA in R do not reproduce Stata’s ordering of numeric missing values, I handled missing observations explicitly and verified the resulting sample counts.. Stata’s time-series operators on a declared panel return missing across gaps in a firm’s history, while R’s lag functions return the previous *observed* row whatever quarter it belongs to. Division by zero returns missing in one language and infinity in the other, and an infinite value passed to a winsorizing routine is capped at the upper percentile and re-enters the regression looking entirely ordinary. None of these produces an error or a warning. The practice that catches them is to write down, for each command without an exact counterpart, what the original does at the edge cases—missing values, gaps, zero denominators, ties—and then test that behaviour directly rather than assume it.

Some of these workarounds are unavoidably approximate. The original’s winsorization performs order-statistic replacement where the reconstruction clips at quantile bounds, and matching the reported percentiles required selecting the order-statistic convention explicitly; the two can still differ by a few observations in the tails. Where an approximation of this kind

is necessary, the useful discipline is to bound its consequences—here, negligible at reporting precision—and record it, rather than leave it implicit.

3. Treat estimation-package defaults as part of the specification. Two commands that implement the same estimator need not report the same numbers. Because estimation defaults can differ across packages and versions, I set the singleton-removal and small-sample-correction options explicitly and recorded the settings used. Neither convention materially moves the point estimates, but both change the reported observation count and the standard errors—exactly the quantities under comparison.

4. When inputs are missing, define in advance what can and cannot be compared. Four of the original’s inputs were unavailable or out of scope, which rules out a substantial share of the published columns. The temptation is to fill the gap: to substitute an available proxy for distance to default, or to quietly drop the affected columns from the comparison. Both make the exercise less informative—the first because the resulting column is no longer the one in the paper, the second because a reader cannot tell whether a column is absent by design or by oversight. I instead fixed the restrictions before estimating anything, and then accounted for every column of every table in one direction or the other: either replicated and compared, or excluded with the specific restriction that removes it. This is what makes a partial replication legible. It also makes the remaining comparisons load-bearing, because the set of things being compared was not chosen after seeing which ones agreed.

5. Look for a published estimate produced under your conditions. The largest gap remaining after the code was corrected—a baseline coefficient of -0.73 against a published -0.57 —had two candidate explanations: a residual error in the translation, or the fact that the shock series available here begins in 1994 rather than 1990. Coefficients alone cannot separate these. An arithmetic check pointed one way, since the shortfall of 48,152 observations is close to sixteen missing quarters at roughly three thousand sample firms per quarter. What settled it was the authors’ own post-1994 subsample table, which estimates the same specification on their data restricted to the window available here: against that benchmark the replication agrees to 0.02 on the coefficient and to within 0.001 on the R^2 . Published robustness tables are free benchmarks, and finding one that matches your conditions converts an unexplained discrepancy into a documented data limitation. It is worth reading the appendix tables of the original for this purpose before assuming a bug.

A related diagnostic is to ask which of your results depend on the missing input and which do not. Quantities identified from the cross-section of firms—lagged investment at

0.203 against a published 0.20, the implied standard deviation of demeaned leverage, the descriptive moments—replicate closely, while every quantity identified from time-series variation in the shocks is larger in magnitude. That the discrepancies sort along that line is itself evidence about their source.

6. Predict what a fix should do before applying it. Each correction was accompanied by a prediction about its effect, which was then checked. Rebuilding the macroeconomic lags to respect panel gaps should have removed observations from the specifications with aggregate controls, since a missing lag drops the row from the regression; it did, moving the sample from 171,250 to 130,745. A fix that changes nothing is a prompt to ask whether it was actually applied—twice in this project a correction appeared to fail because a superseded block of code was still in the script above its replacement, so both versions ran. Rebuilding the pipeline from the top after editing a construction step, and asserting the invariant immediately before the operation that depends on it, are cheap insurance against this.

Finally, not every difference is an error, and not every error is worth correcting. A control listed both as a regressor and as an absorbed fixed effect in the original is dropped as collinear by Stata and is correctly omitted rather than missing from the R specification; the two packages use different small-sample conventions for instrumental-variables standard errors, which cannot be reconciled by editing a specification; and the winsorization routines differ by a few tail observations at reporting precision. Each was identified, understood and left in place. Documentation is a legitimate resolution, and treating it as one avoids the greater risk of going down the rabbit hole editing a correct translation until it reproduces a target number.

9 Conclusion

This paper documents a restricted replication of the empirical results in Ottonello and Winberry (2020), reconstructing the authors’ Stata pipeline in R. Within the agreed scope — leverage interacted with the baseline wide-window monetary shock — the replication reproduces the published results. Benchmarked against the authors’ own post-1994 estimates, which match the sample coverage available here, the baseline interaction coefficient is -0.73 (s.e. 0.35) against a published -0.71 (s.e. 0.35), with the R^2 agreeing to within 0.001 and the sample within 1.7%. Across the robustness tables, every in-scope coefficient lies within one standard error of its published counterpart and reproduces the direction and approximate magnitude of the adjustment relative to the baseline. Two results replicate only partially: the relative size of the expansionary and contractionary coefficients in Table XI, where nei-

ther estimate is significant, and the amplitude of the local projection in Figure 1(a), which is sensitive to sample composition within the post-1994 window. Every excluded column across the paper’s tables is accounted for by one of five documented data restrictions or by the exclusion of the structural model from the project’s scope.

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